

Single - Pendulum MPC

The pendulum has a state, and takes an input control $[\tau]$ in order to make changes to the state.

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

State

$$u = [\tau]$$

control input

We also need to define a target state, where we aim at achieving stabilisation

$$x_{ref} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \text{Upright position with zero velocity}$$

MPC is basically a cost calculator $\Rightarrow$ It / We have to formulate an objective cost J^* which we aim at minimizing for optimal trajectory ideally.

Objective $\Rightarrow$ we need to reach the final state, and the cost function caters to that.

Cost Function

$$\text{Cost} = \sum_{k=0}^{N-1} (x_k - x_{\text{ref}})^T Q (x_k - x_{\text{ref}}) + u_k^T R u_k \\ + \text{Terminal cost}$$

$(x_k - x_{\text{ref}}) \Rightarrow$ This is basically the Δ we get when comparing current to the reference state $\Rightarrow$ Error

Error $\propto$ Cost $A[\text{error}]$ is a (2×1) matrix

$\Rightarrow Q$ is 2×2 in nature

$Q \Rightarrow$ State penalty

$\Rightarrow$ How far it is from where it has to be at

S^{th} : $[R]$ is $[1 \times 1]$ in nature, and imposes penalty on how strongly the motor / s is / are actuated.

Terminal cost: $(x_N - x_{\text{ref}})^T P (x_N - x_{\text{ref}})$

$\hookrightarrow$ This is relatively a stronger penalty, because we want our system to actually reach the end state [at least in this scenario]

$P \rightarrow (2 \times 2)$ in nature

This cost function cumulates over one horizon, with N steps which are discretized in nature (naturally); ~~and~~

Constraints

$x_0 = x_{\text{sensor}} \rightarrow$ At the start of a timeline

$|u| \leq T_{\text{max}} \rightarrow$ Physical limit of motor

$x_{k+1} = x_{\text{next}} \rightarrow$ Constrained environment set up where the final torques are computed (via RK4)

The equation of motion is continuous in nature $\rightarrow$ but we need discreteness in order for our system to work $\Rightarrow$ We use RK4

Say we have $\dot{x} = f(x, u)$

Current: x_k, u_k

P.T.O

4 eqns:

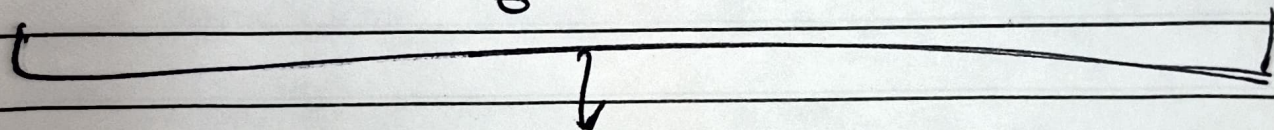
$$k_1 = f(x_k, u_k)$$

$$k_2 = f\left(x_k + \frac{dt}{2} k_1, u_k\right)$$

$$k_3 = f\left(x_k + \frac{dt}{2} k_2, u_k\right)$$

$$k_4 = f\left(x_k + dt \cdot k_3, u_k\right)$$

$$x_{next} = x_k + \frac{dt}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$



This is the constraint that has been set-up.